MSD MANUAL
Consumer Version

Fever in Adults

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Fever is an elevated body temperature that occurs when the body's thermostat (located in the hypothalamus in the brain) resets at a higher temperature, primarily in response to an infection. Elevated body temperature that is not caused by a resetting of the temperature set point is called hyperthermia.

Although 98.6° F (37° C) is considered normal temperature, body temperature varies throughout the day. It is lowest in the early morning and highest in the late afternoon—sometimes reaching 99.9° F (37.7° C). Similarly, a fever does not stay at a constant temperature. Sometimes temperature peaks every day and then returns to normal. This process is called intermittent fever. Alternatively, temperature varies but does not return to normal. This process is called remittent fever. Doctors no longer think that the pattern of the rise and fall of fever is very important in the diagnosis of certain disorders.

Body temperature can be measured at several areas of the body. The most common sites are the mouth (oral temperature) and rectum (rectal temperature). Other sites include the ear, forehead, and, much less preferably, armpit. Temperature is typically measured using a digital thermometer. Glass thermometers containing mercury are not recommended because they can break and expose people to mercury.

Oral temperatures are considered elevated when

- They are higher than 99° F (37° C) in the early morning.
- They are higher than 100.4° F (38° C) at any time after the early morning.
- They are higher than a person’s known normal everyday temperature.

Rectal and ear temperatures are about 1.0° F (0.6° C) *higher* than oral temperatures.

Skin temperatures (for example, the forehead) are about 1.0° F (0.6° C) *lower* than oral temperatures.

Many people use the term “fever” loosely, often meaning that they feel too warm, too cold, or sweaty, but they have not actually measured their temperature.

The ability to generate a fever is reduced in certain people (for example, those who are very old, very young, or who have an alcohol use disorder).

(See also Overview of Infectious Disease and Fever in Infants and Children.)

Consequences of fever

The symptoms people have are due mainly to the condition causing the fever rather than the fever itself.

Although many people worry that fever can cause harm, the typical temporary elevations in body temperature ranging from 100.4° to 104° F (38° to 40° C) caused by most short-lived (acute) infections are well-tolerated by healthy adults. As one of the body's defenses against an infection, a fever can trigger the production of antibodies and make it more difficult for microorganisms to grow, and may lessen the severity of an infection and help it go away.

However, a moderate fever may be slightly dangerous for adults with a heart or lung disorder because fever causes the heart rate and breathing rate to increase. Fever can also worsen mental status in people with dementia.

Additionally, an extreme temperature elevation (typically more than 105.8° F, or 41° C) may be damaging. A body temperature this high can cause malfunction and ultimately failure of most organs. Such extreme elevation sometimes results from very severe infection (such as sepsis, malaria, or meningitis) but is more typically caused by heatstroke or use of certain illicit drugs (such as cocaine or PCP).

Fever in healthy children can cause febrile seizures.

Causes of Fever in Adults

Substances that cause fever are called pyrogens. Pyrogens can come from inside or outside the body. Microorganisms and the substances they produce (such as toxins) are examples of pyrogens formed outside the body. Pyrogens formed inside the body are usually produced by 2 types of white blood cells called monocytes and macrophages. Pyrogens from outside the body can cause fever by stimulating the body to release its own pyrogens or by directly affecting the area of the brain that controls temperature.

Many disorders can cause fever. They are broadly categorized as

- Infectious (most common)
- Neoplastic (cancer)
- Inflammatory (including autoimmune disorders, allergic reactions, and some drug reactions)

An infectious cause is highly likely in adults with a fever that lasts 4 days or less (called an acute fever). An acute fever in people with cancer or a known inflammatory disorder also is most likely to have an infectious cause. In healthy people, an acute fever is unlikely to be the first sign of a chronic illness. A noninfectious cause is more likely to cause fever that lasts a long time or returns.

Infectious causes

Infections are the most common causes of fever. Infections can be caused by bacteria, viruses, or fungi. Virtually all infections can cause fever, but, overall, the most likely infectious causes are the following:

- Upper and lower respiratory tract infections
- Gastrointestinal infections
- Urinary tract infections (UTIs)
- Skin infections

Most acute respiratory tract and gastrointestinal infections are viral.

Neoplastic causes

Many cancers cause fever, for example, leukemia, lymphoma, and kidney cancer.

Inflammatory causes

Inflammatory disorders that cause fever include systemic rheumatic disorders such as rheumatoid arthritis, systemic lupus erythematosus (lupus), and giant cell arteritis.

Fever may also result from an allergic reaction.

Medications and drugs sometimes cause fever. For example, beta-lactam antibiotics (such as penicillin) and sulfa medications can trigger a fever. Certain illicit drugs (such as cocaine, amphetamines, or phencyclidine) and medications such as anesthetics and antipsychotics can cause an extremely high temperature.

Risk factors

Certain factors help doctors determine what is the most likely cause of fever in a person. These factors include the following:

- The person's health status, age, and occupation
- Hospitalization
- Use of certain medications or illicit drugs
- Exposure to infections (for example, through travel or contact with infected people, animals, or insects)

Some Causes of Fever Based on Risk Factors

Risk Factor	Cause
None (healthy)	Upper or lower respiratory tract infection Gastrointestinal infection Urinary tract infection Skin infection
Hospitalization	Bloodstream infection related to a catheter inserted in a vein (IV catheter infection) Urinary tract infection, particularly in people with a urinary catheter Pneumonia, particularly in people on a ventilator Atelectasis (collapse of part of a lung due to an airway blockage, rather than an infection) Infection or a pocket of blood (hematoma) at the site of surgery Deep vein thrombosis or pulmonary embolism Diarrhea (due to *Clostridioides difficile*–induced colitis) Medications Transfusion reaction Pressure sores
Travel to areas where an infection is common (endemic areas)	Malaria Viral hepatitis Disorders that cause diarrhea Typhoid fever